# IncobotulinumtoxinA for Aesthetic Treatments: Review and Recommendations From Latin America

Carla de Sanctis Pecora, MD,* Ada Trindade de Almeida, MD,† Vinicius Figueredo, MD,‡ Cyro Hirano, MD,§ Alejandra Bugallo, MD,‖ Cristina Cortés, MD,¶ Esperanza C. Welsh, MD,** Julieta Spada, MD,†† Nicolás Heredia, MD,‡‡ Victoria de La Fuente, MD,§§ and Mariana Muniz, MD‖‖

**BACKGROUND** As new aesthetic treatment techniques are developed, the understanding of botulinum toxin type A safety and treatment outcomes continues to evolve.

**OBJECTIVE** This article was developed to provide a comprehensive up-to-date reference for clinicians using incobotulinumtoxinA (INCO; Xeomin/Bocouture, Merz Pharmaceuticals GmbH, Frankfurt, Germany) for aesthetic treatments.

**PATIENTS AND METHODS/MATERIALS** A Latin American panel of 11 physicians was assembled to discuss and develop recommendations on the use of INCO for aesthetic treatment based on the literature review and their private medical practice.

**RESULTS** The panel found that INCO is a highly purified botulinum toxin, without other clostridial proteins (hemagglutinins and nonhemagglutinins). IncobotulinumtoxinA has proven its efficacy and duration over time, maintaining response after multiple treatment cycles.

**CONCLUSION** This recommendation provides up-to-date information on the use of INCO for aesthetic treatment, focusing on the differential properties of this product regarding its purity, performance, customization, and patient satisfaction.

Since its first approval by the US FDA in 1989, botulinum toxin A (BoNTA) treatment has become a useful and popular procedure in the aesthetic field. The number of new therapeutic and aesthetic indications for BoNTA has considerably increased, along with the number of injection techniques and the authors' understanding of the safety profile.[1,2]

While the number of commercially available BoNTA products is increasing in Latin America and worldwide, the 3 best-known formulations are abobotulinumtoxinA (Dysport/Azzalure, Ipsen Biopharm), incobotulinumtoxinA (INCO; Xeomin/Bocouture, NT 201; Merz Pharmaceuticals GmbH, Frankfurt, Germany), and onabotulinumtoxinA (ONA; Botox/Vistabel, Allergan, Inc.). Although these products contain the same type A serotype, different manufacturing and stabilization processes produce formulations containing varying amounts of neurotoxin-associated complexing protein. Therefore, these BoNTAs have different molecular weights, three-dimensional structures, antigenicity risks, biological activity, pH, and storage requirements. IncobotulinumtoxinA, a highly purified BoNTA, was first approved in Latin America in 2008 and remains the only formulation specially designed to contain only the active neurotoxin molecule, free of complexing proteins.[1,3]

To date, there are no Latin American recommendations on injection techniques or optimal INCO doses for specific aesthetic indications.[4–7] A panel of 11 dermatologists and plastic surgeons from Latin America (Argentina, Brazil, Colombia, and Mexico)—each with more than 10 years of experience in the aesthetic use of INCO—was assembled to review the current medical literature and compose a structured guideline on the use of INCO for aesthetic treatment.

This manuscript focuses on 3 chemical and clinical characteristics of INCO in scientific literature: purity, performance, and customization.

## Purity

The 3 main commercial BoNTA products (Botox, Dysport, and Xeomin) contain toxins from a *Clostridium botulinum*

*From the* Private Practice, Clínica Dermatologie, São Paulo, Brazil; † Private Practice and Hospital do Servidor Público Municipal de São Paulo, Brazil; ‡ Private Practice, Vinicius Figueredo Dermatology, São Paulo, Brazil; § Private Practice, CD Clínica Dermatológica, Rio de Janeiro, Brazil; ‖ Private Practice, Bugallo Centro de Estetica y Laser, Buenos Aires, Argentina; ¶ Private Practice, Escuela de Medicina y Ciencias de la Salud del Tec de Monterrey, Monterrey, México; ** Welsh Derm, Monterrey Private Office, Monterrey, Nuevo Leon, Mexico; †† Private Practice, Spada Dermatología y Estética, Buenos Aires, Argentina; ‡‡ Private Practice, Face and Nose Institute, Bogota, Colombia; §§ Private Practice, Levels High-End Dermatology Clinic, Ciudad de México, Mexico; ‖‖ Private Practice, Mariana Muniz Dermatology, São Paulo, Brazil

*All authors report grants as consultant and have been speakers for Merz Pharmaceuticals.*

*Merz Aesthetics LATAM funded medical writing assistance and expert advisory board. Despite this support, the discussions and this manuscript were conducted with autonomy and independence by the medical authors.*

*Address correspondence and reprint requests to: Carla de Sanctis Pecora, Clinica Dermatologie, Avenida Dr. Cardoso de Melo, 1308, conj. 121, Vila Olimpia, São Paulo 04548-004, Brazil, or e-mail: carla@dermatologie.com.br*

*© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. All rights reserved.*

*Dermatol Surg 2024;50:S24–S34*
*http://dx.doi.org/10.1097/DSS.0000000000004343*



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

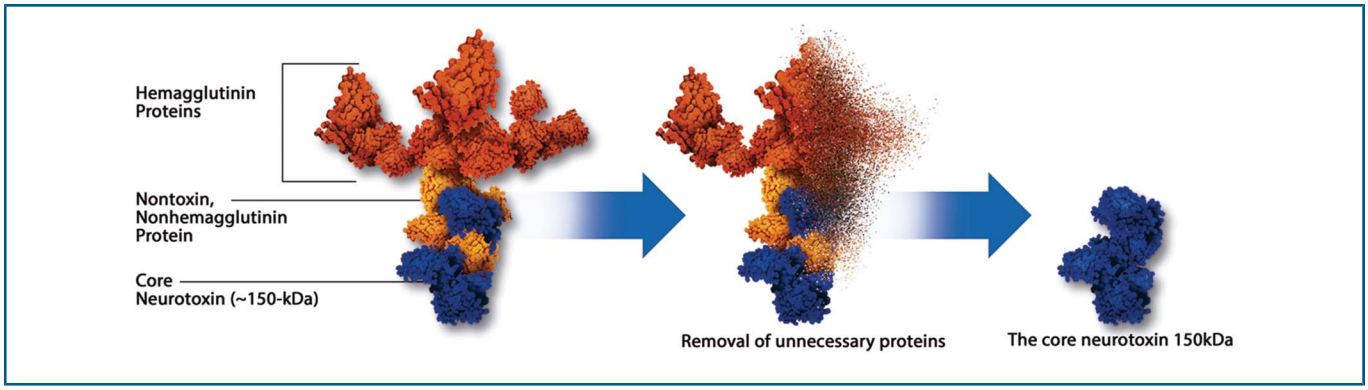



**Figure 1.** Complex BoNTA protein and stepwise chromatographic removal of all complexing proteins, resulting in the 150-kDa pure neurotoxin. As a bacterial protein, BoNTA is foreign; however, no known T-lymphocyte pattern recognition receptors exist for the 150-kDa active neurotoxin. In nature, the immune system does not encounter pure neurotoxin and therefore only recognizes the bacteria that produce the toxin. DCs do not recognize pure neurotoxin as foreign or dangerous since it is formulated without complexing proteins; there is no antigen presentation or activation of the T lymphocytes. BoNTA, botulinum toxin type A; DCs, dendritic cells.

Hall strain, which is produced in nature as part of a larger multimeric complex with accessory proteins.[1,3,8] As a result of the exclusive double purification manufacturing process that removes complexing proteins, the INCO formulation contains only the 150-kDa active neurotoxin responsible for the therapeutic effect (Figure 1). The purification process is performed in a series of chromatographic steps to minimize the risk of inactive toxin content and thus limit possible denaturation. Sucrose and human serum albumin are added (as inert excipients) for the final product stabilization, minimizing the risk of neurotoxin inactivation during manufacturing and long-term storage.[1,8,9] Stability is important because if the 150-kDa active molecule is inactivated, although it will not be taken up by nerve cells, it may be recognized by the immune system, thus increasing immunogenicity.[1,3] Furthermore, it has been demonstrated that the stability of the BoNTA complex is pH dependent. In other formulations, the complexing proteins dissociate from the active neurotoxin with a half-life of <1 minute after reconstitution under physiologic conditions, indicating that these accessory proteins are not necessary for the diffusion or stability of the active neurotoxin.[1,8]

Apart from the neurotoxin complex, manufacturing processes may also contribute to the presence of other bacterial elements in product formulations. Bacterial byproducts such as hemagglutinin, flagellin, and *C. botulinum* DNA fragments can be recognized by specific pattern receptors (TLR9, TLR5) on dendritic cells (DCs) as foreign and dangerous, activating the immune cascade in a similar manner to adjuvant activation in vaccination.[2,10] To date, there is no specific pattern recognition receptor for the 150-kDa active neurotoxin; however, once activated and differentiated, DCs are able to recognize all proteins surrounding them, even the pure neurotoxin.[10–12] Purity is associated with best biological product manufacturing practices and reduced risk of hypersensitivity. This is especially true after multiple BoNTA treatments, where repeated exposure to antigenic proteins can cause an increased risk of immunogenicity.[9,13]

Total BoNTA dosage can often reach the range used for therapeutic indications; facial treatment with BoNTA commonly includes 100 units (U) per session.[10,14] With increased exposure comes a greater risk of developing secondary nonresponsiveness to BoNTA treatment. Several factors are associated with secondary nonresponsiveness, such as incorrect toxin injection technique, inadequate dosage, individual variation in response, storage, handling, and even inadequate product reconstitution.[15–17] Repeated injections of BoNTA, total dose per session >200 U with a cumulative dose >1500 U, and application intervals <3-months may lead to partial secondary treatment failure (PSTF), due to the development of neutralizing antibodies (NABs).[18] Therefore, it is important to be aware of the first clinical signs of PSTF, including an increase in required dose and variable results over the duration of treatment.[19,20]

In a clinical study, switching from other BoNTAs to INCO demonstrated a decline in the titer of NABs in patients with cervical dystonia and PSTF over a period of ≤50 months.[15,21] INCO has been proven safe for the treatment of patients with cervical dystonia, using flexible doses and application intervals (≤12 weeks, 6 weeks minimum) and without development of NABs after 3 years of follow-up.[19,20] Clinical studies have also shown that patients treated exclusively with INCO did not develop NABs or present PSTF, even with doses of up to 800 U for the treatment of upper limb spasticity.[8] The authors suggest the use of highly purified INCO to minimize the risk of immunogenicity and allow for long-term use.[2,3]

## Performance

IncobotulinumtoxinA has proven its efficacy and duration over time and after multiple treatments.[8] Although units of different BoNTA formulations are not interchangeable, evidence from the literature has demonstrated that ONA and INCO have a clinical equivalency ratio of 1:1 (1 U of INCO is equivalent to 1 U of ONA).[1,22] As with other BoNTA products, INCO's effect and duration increase



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

**TABLE 1. Recommendations of Latin American Experts for INCO Treatment of the Main Facial Muscles**

| Muscle Group | Action | Criteria for Treatment | Points of Injection | Dosage | Injection Technique* |
|---|---|---|---|---|---|
| Occipito-frontalis | Moves de scalp back, raises the eyebrow, and wrinkles the forehead | Forehead wrinkles (horizontal lines in the forehead) at rest, and/or moderate-to-severe dynamic wrinkles | 6–21 | 0.5–2 U/point | Insert 30% of the needle angled upward. Extend the injection sites laterally along the lateral canthus |
| Corrugator supercilii | Pulls the eyebrows downward and toward the midline | Patient willingness to reduce frequent glabellar contractions transmitting not real negative messages. Vertical medial lines in the glabellar region | 2 (origin and tail)<br>*3rd point (lateral, blending with orbicularis oculi fibers) | 4–6 U/point<br>*3rd point: 0.5–2 U | Origin: introduce 100% of the needle<br>Insertion: introduce 30% of the needle<br>*3rd point: introduce the needle bevel only |
| Procerus | Draw down the skin between the eyebrows | Horizontal lines located at the root of the nose | 1–2 | 4–8 U | Introduce 50% of the needle directed upward |
| Depressor supercilii | Draws down the head of the eyebrows | Diagonal lines medially to the medial end of the eyebrow | 1 | 1–2 U | Subdermal injection, directing the needle medially and upward |
| Orbicularis oculi | Responsible for closing the eyelids, blinking, and developing crow feet | Periorbital wrinkles in contraction and at rest, and pretarsal hypertrophy. A drop of the lateral brow | 3–6 | 2–4 U/point (lateral radial lines)<br>1–2 U/point (inferior line and lateral end of the brow) | Insert 30% of the needle superficially, directing the needle away from the eye |
| Nasalis | Compresses the nostrils and may completely close them | Asymmetry of nasal wrinkles when sniffing | 1–2 | 2–4 U/point | Intramuscular injection with the needle at 45° with the skin, on the superior third of the lateral nasal wall, inferior to the angular vein |
| Levator labii superioris alaeque nasi | Dilates the nostril and elevates the central-lateral upper lip, widens and raises the nostril wings, and pulls the infraorbital triangle upward. Also contributes to nasal wrinkles | Bunny lines (oblique lines on the nose sides) pull the center of the upper lip up. Horizontal wrinkle above the upper lip. Deepening of the nasolabial fold. Contributes to gingival smile | 1 | 0.5–2 U | Upper lateral nasal wall and/or near the nasal ala. Insert only 50% of the needle pointed upward |
| Levator labii superioris | Elevates and everts the upper lip | Deepens the nasolabial fold and raises the upper part of it. Raises the central and lateral part of the upper lip. | 1 | 1–2 U | Injection with needle perpendicular to the skin, lateral to the bony nasal prominence. Insert |



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

**TABLE 1. Recommendations of Latin American Experts for INCO Treatment of the Main Facial Muscles *(Continued)***

| Muscle Group | Action | Criteria for Treatment | Points of Injection | Dosage | Injection Technique* |
|---|---|---|---|---|---|
| | | Exposure of the superior dental arch while smiling (frequently involved in the gingival smiling) | | | 50% of the needle pointed upward |
| Pre Tarsal orbicularis oculi | Pulls both the upper and lower eyelids and some of the adjacent skin below the eye together and toward the inner corner of the eye, narrows the eye aperture, the lower lid becomes more straight and bulging | Asymmetry of the eye aperture, eyelid ptosis, bulge, and dynamic wrinkles in the inferior and medial eyelid | 1–5 points | 0.5–1 U/point | Intradermal injection 3 mm below or above the ciliary margin of the related inferior or superior eyelid |
| Levator anguli oris | Pulls the angle of the mouth straight up and curves the mouth line upward at its ends | For the most part, it is not used in basic emotions, rather primarily it stabilizes the modiolus | No consensus | No consensus | No consensus |
| Zygomaticus major | Pulls the angle of the mouth up out and backward when smiling | Pulls the corners of the lip up, backward, and outward. Deepens the nasolabial fold, pulling it laterally and up | 1–2 | 1–2 U/point | Superficial injection (50% of the needle, perpendicular to the skin): 1st point 2 cm from the oral commissure on a diagonal line to the malar prominence. 2nd point on the malar prominence |
| Zygomaticus minor | Pulls the middle section of the nasolabial furrow and the middle portion of the side of the upper lip outward and slightly upward | Lip pulled upward and laterally, deepening the upper middle portion of the nasolabial furrow | 1 | 0.5–2 U | Superficial injection (50% of the needle, perpendicular to the skin) 1 cm above the zygomaticus major |
| Orbicularis oris | Kissing, puckering, and pressing lips against the teeth, closing the mouth | Perioral wrinkles | 2–8 | 0.5 U/point | Superficial injection at or 2 mm above the vermilion border. Insert only the needle bevel pointed upward |
| Risorius | Pulls the modiolus and therefore the angle of the mouth out and backward | Lateral retraction of the angle of the mouth (sardonic expression) | 1 | 1–2 U/point | Deep injection (100% of the needle, perpendicular to the skin) 2 cm medially to the anterior border of the masseter |



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

**TABLE 1. Recommendations of Latin American Experts for INCO Treatment of the Main Facial Muscles *(Continued)***

| Muscle Group | Action | Criteria for Treatment | Points of Injection | Dosage | Injection Technique* |
|---|---|---|---|---|---|
| Mentalis | Raises and wrinkles the skin of the chin, thus elevating the lower lip | "Orange peel" texture and deep transverse mental crease | 1–2 | 2–4 U/point | Deep injection (100% of the needle, perpendicular to the skin) on the chin lower, inferior to the midline, at a distance >2 cm from the lower lip |
| Depressor labii inferioris | Pulls the middle third of the lower lip straight downward | Pulls the lower lip down. Exposure of lower teeth arch smiling | 1–2 | 0.5–1 U/point | Superficial injection (30%–50% of the needle) 1–1.5 cm from the lower lip border, 1 cm from the midline |
| Depressor anguli oris | Depresses and lateral displace the angle of the mouth | The corner of the lips pulled down, deepening the labial mentalis furrow (not nasolabial) | 1 | 2–4 U | Superficial injection (50% of the needle, perpendicular to the skin) immediately above the angle of the mandible and 1 cm lateral to the oral commissure |
| Platysma (bands and jawline) | Pulls the lower lip and corner of the mouth sideways and downwards | Deep vertical smiles line lateral to the modiolus, and sometimes horizontal lines below the medial lower lip. Pulls the lower portion of the nasolabial fold. May produce horizontal lines on the neck | 6–10 | 2 U/point | Superficial intramuscular injections, 1.5–2 cm apart along the mandible margin and platysma bands |

INCO, incobotulinumtoxinA.
* Injection technique has been taken into account using a 6-mm needle

nonlinearly as long as higher doses of neurotoxin reach each neuromuscular junction.[23]

Treatment efficacy is also dependent on correct reconstitution of the product.[17] In Latin America, INCO is commercialized as Xeomin in a 100 U vial supplied in a lyophilized (freeze-dried) form that must be reconstituted in a preservative-free sterile saline solution before injection. The authors recommend to reconstitute a 100 U vial of INCO in 2 mL of sterile saline solution (1 U corresponds to 0.02 mL of the final solution). It is important to swirl and invert the vial, mixing the solution thoroughly to ensure that the powder has been evenly dissolved.

Sustainability is an important factor during INCO's production process. No animal testing is performed, and manufacturing and transportation of INCO produces less carbon dioxide emissions than other BoNTA formulations since it does not need to be kept cold. In addition, the excess secondary packaging required for cold chain supply management not only imposes a larger environmental impact but also represents a greater burden for hospital pharmacy systems regarding storage space, refrigerated storage capacity, and disposal.[24]

## Customization

Understanding an individual's muscle anatomy and contraction pattern is a key element in determining the appropriate neuromodulator treatment for dynamic lines; studies have shown that motor endplate-targeted injections optimize the effect of BoNTAs,[25] and high-density surface electromyography has demonstrated how variable and diffusely placed motor endplates can be in the frontalis muscle.[26] Careful assessment should include individual anatomy, muscle mass, strength and function, expectations, movement habits, gender, ethnicity, and age[27,28] (Table 1).



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

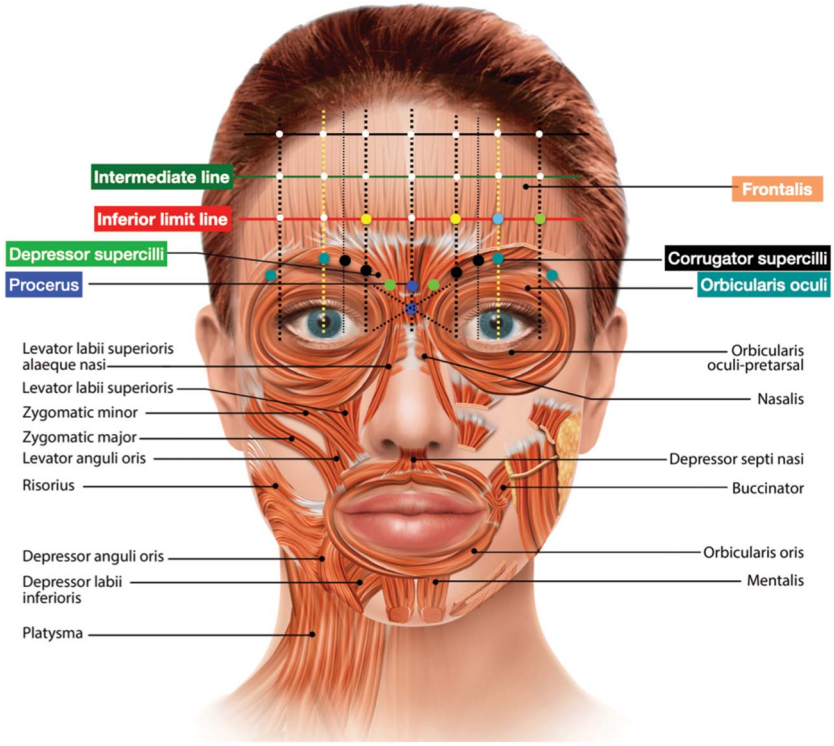



**Figure 2.** Schematic mapping of the ONE21 technique for treating the upper third of the face, including the *procerus* (royal blue dots), *corrugators* (black dots), *depressor supercilii* (green dots between eyebrows), *orbicularis oculi* (teal dots), and the *frontalis*. Treatment of the *frontalis* can be further described for forehead lines and eyebrow reshaping. *Forehead lines:* the assessment grid results in 21 potential injection sites defined by 7 vertical lines and 3 horizontal lines. Vertical lines include 2 midpupillary lines (yellow dash); 5 lines based on the inner and outer canthus and the medial facial line (black dash). Horizontal lines include inferior limit line (red solid; lowest forehead line), upper line (black solid; highest forehead line), and intermediate line (green solid). The injection points and dosage are based on the palpation of the muscle contraction. *Eyebrow reshaping:* injection in the inferior limit line—lateral arching effect (yellow dots); flattening effect (light blue dot); dropping of the lateral end of the brow (green dot on the inferior limit line).

Treating all patients with the same injection pattern and doses can result in insufficient or undesired outcomes.[27,28]

## Forehead

The ONE21 technique is a customizable injection protocol that allows an individualized assessment and treatment of the upper third of the face with neurotoxin, leading to a predictable eyebrow shape.[29] This protocol was further improved to also include the treatment of glabellar lines.

The final assessment of the forehead determines a grid with 21 intersecting points corresponding to potential injection sites (Figure 2) based on 7 vertical lines (anatomical landmarks) and 3 horizontal frown lines (functional lines). The ONE21 technique promotes wide coverage of all neuromuscular junctions across the *frontalis* by increasing the number of potential injection sites while decreasing the dose per injection point.

The injection points are defined by the wrinkles' shape and position (visual assessment)[30,31] and the presence or absence of muscle contraction, while the dosage at each point is based on muscle strength during contraction (0—no muscle contraction, 0.5 U—weak, 1 U—moderate, or

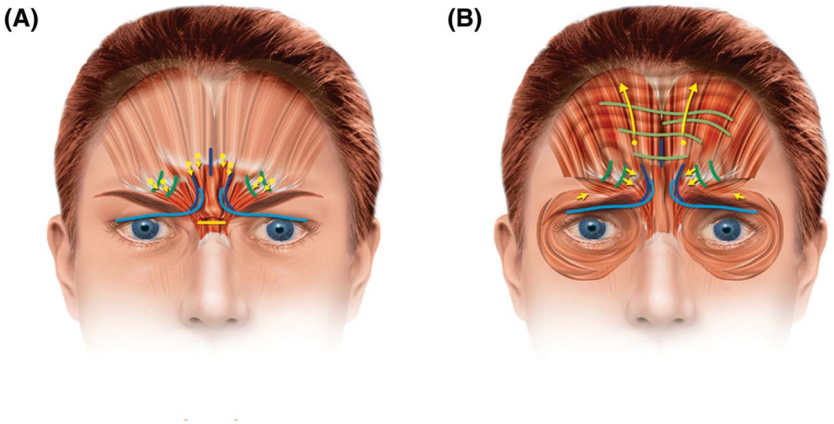



**Figure 3.** Patterns of the glabellar wrinkles according to the muscles involved in the contraction. (A) V-shaped contraction pattern: dark blue line: *corrugator supercilii*; yellow line: *procerus*; light blue line: *depressor supercilii*; green lines: combined action of the orbicular. (B) Omega-pattern: dark blue line—*corrugator supercilii*; green lines—combined action of the orbicular; green light parallel lines—*frontalis*.



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

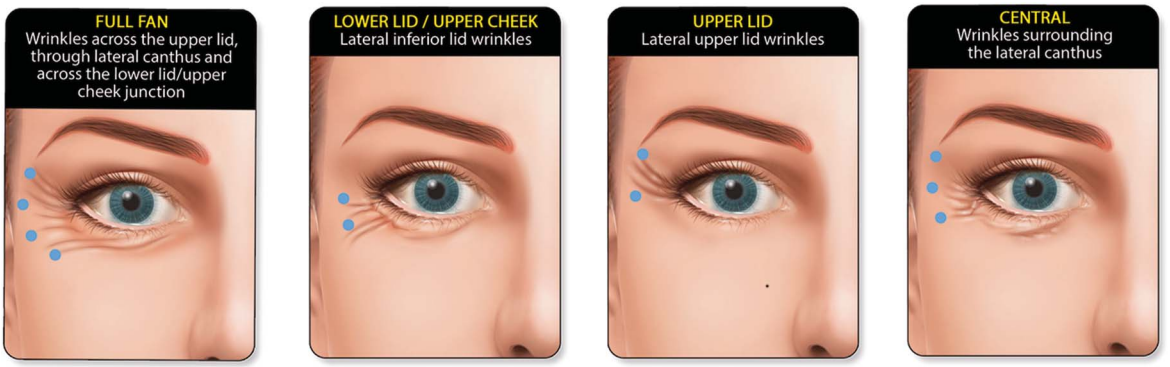



**Figure 4.** Crow feet lines are classified into 4 patterns with suggested injection points.

2 U—intense). For male patients, doses can be increased by 1 U. The injections must be placed intramuscularly, with the needle at a 90°.[27,29] Total dosages range from 3 to 42 U.

To reshape the eyebrows, injections should be administered at the lowest horizontal forehead line (inferior limit line) (Figure 2). To arch and laterally raise the eyebrow, injections should be medial to the midpupillary line (yellow dots); to flatten the eyebrow, injections should be administered at the intersecting point with the midpupillary line (light blue dot); and to drop the eyebrow, injections should be lateral to the midpupillary line (green dot on the inferior limit line). To achieve a soft arching effect of the eyebrow, a decreasing dosage gradient should be injected in the medial-to-lateral direction (Figure 2). To maintain the eyebrow shape, equal doses should be injected at all points along the inferior limit line. For male patients, more units should be injected laterally to keep the eyebrow flattened.[27,29]

## Glabella

Several muscles are involved in the contraction of the glabella: *procerus*, *corrugator*, *depressor supercilii*, and *orbicularis oculi*, all of which intersect with *frontalis* fibers. In addition, unlike the *frontalis*, the depth of the *medial depressor* muscles varies considerably. The combined effects of these muscles can result in different patterns of dynamic lines or asymmetrical wrinkling.[27,28,32,33] There are 5 main glabella contraction patterns: U, V, converging arrows, omega, and inverted omega (Figure 3).[34] These classifications highlighted individual differences in the dominant muscle contractions causing glabellar lines.

The ONE21 technique for the glabella is based on each patient's pattern of wrinkle shape and position, according to the premise that every wrinkle is formed perpendicularly to the muscle fibers that are contracting (Figure 2). In general, vertical medial lines in the glabella are related to the contraction of the *corrugator* muscle (Figure 2; black dots). Horizontal lines at the root of the nose result from contraction of the *procerus* (dark blue dots), and horizontal parallel lines in the central part of the forehead are related to a combined action of the *frontalis*. The presence of diagonal lines medially to the medial end of the eyebrow denotes the action of the *depressor supercilii* (green dots between the eyebrows). Vertical parallel lines laterally above the brows indicate the intersection of the *corrugator* tail fibers with the *orbicularis oculi* (teal dots). The final ONE21 assessment results in a highly customized injection protocol that may vary from 3 to 12 injection points (Figure 2).[33] The physician panel agreed on an optimal starting dose of 20 U for female patients and 30 U for male patients, to be adjusted to the individual's needs.

## Periocular

Crow feet lines resulting from the contraction of the *orbicularis oculi* in the lateral canthus region are one of




**Figure 5.** The inferior ocular wrinkles classification and suggested injection points.



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

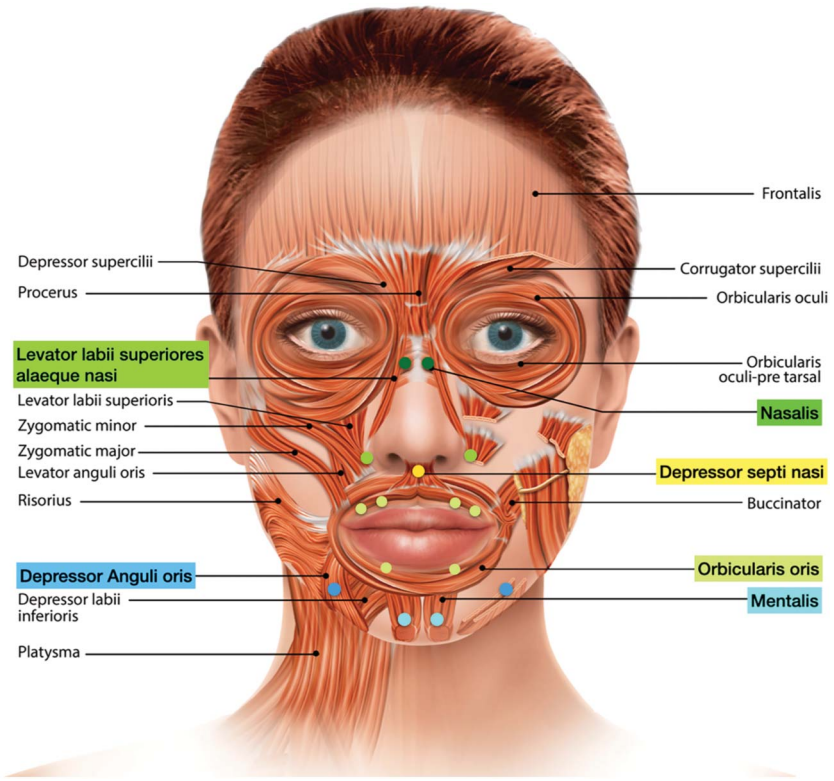



**Figure 6.** Injection points for the treatment of the mid and lower third of the face. *Middle third: nasalis* (dark green dots) and *depressor septi nasi* (yellow dot). *Lower third:* for gummy smile, treat the *levator labii superioris alaeque nasi* (lime green dots); for barcode wrinkles, treat the *orbicular oris* (pale green dots around lips); for treatment of hyperactivity of the *mentalis* (light blue dots) and the *depressor anguli oris* (dark blue dots).

the most common complaints of patients. They are classified into 4 different patterns (Figure 4): central fan (rhytids limited to the lateral canthus region), full fan (rhytids in the upper, lateral, and lower region), lower fan (rhytids in the lower part), and upper fan (rhytids in the upper part).[35]

Injection should be in the superficial plane, with the needle angled laterally away from the *orbicularis oculi*. The first injection point should be placed at 1.5 to 2 cm in line with the lateral canthus and 2 others 1 to 1.5 cm above and below the first, at an approximate 30° angle medially (Figure 4; blue dots) for a total of ~12 U per side. For lateral elevation of the eyebrow, the technique should be complemented with the injection of 1 to 3 U of INCO about 1 cm from the orbital rim to block the superior lateral fibers of the *orbicularis oculi*.

Inferior ocular wrinkles have been categorized into 3 types: B1 (inferior lateral lines), B2 (inferior medial lines), and B3 (inferior medial lines). For each type, a single point of intradermal INCO injection of 0.5 to 1 U is recommended (Figure 5; blue dots). For the treatment of type B1 lines, the injection point must be placed 1 cm below and aligned with the lateral canthus; for type B2 lines, 3 to 5 mm from the border of the inferior lid in line with the midpupillary line; and for type B3 lines, 5 mm below the inner canthus (Figure 5).[36] Before treating inferior ocular wrinkles, a careful assessment should verify the presence of visible adipose tissues, flaccidity (snap test) of the skin, and/or the tarsus portion of the muscle, and whether the patient complains of edema in the region. The presence of any of these conditions is a contraindication for INCO injection.[36]

## Perinasal Area

The anatomy of the mid and lower face is complex, with many proximal and overlapping muscles. For nose wrinkling ("bunny lines"), injections should be placed in each side of the *nasalis* muscle, with a dose of 1 to 3 U of INCO per side (Figure 6; dark green dots). For nasal tip ptosis, the *depressor septi nasi* (yellow dot) can be treated. The injection point is placed in the columella, and the dose ranges between 2 and 4 U. For patients with a gummy smile, the *levator labii superioris alaeque nasi* muscle is usually hyperkinetic. It can be treated with an INCO injection of 1 to 3 U placed in each lateral nasal side wall (Figure 6; lime green dots).

## Perioral Area

The *orbicularis oris* muscle is a circular, sphincteric muscle important for several lip functions, such as expressing emotions, sealing the mouth, speaking, blowing, and kissing. Owing to age-related skin atrophy, primarily in women, the repeated contraction of this muscle forms unsightly vertical wrinkles around the lips, known as "barcode" wrinkles. The superficial application of INCO above the vermilion border reduces these wrinkles and can evert the lips, giving the impression of increased volume (Figure 6; pale green dots). Doses should be minimal, varying from 0.5 to 2 U per lip quadrant for aesthetic



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

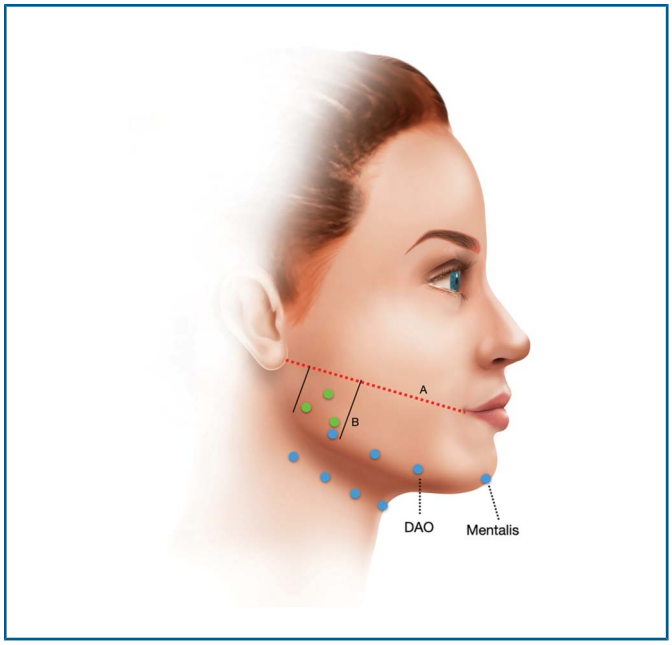



**Figure 7.** Injection points for treating *masseter* hypertrophy and facial contouring. *Masseter:* the safety zone for application should be determined by drawing a line connecting the earlobe to the corner of the mouth (A: red dotted line), while the mandible defines its lower border. The injection points (green dots) should be located below this line, in area B (black lines), which defines the anterior border of the *masseter*. *Facial contouring:* injection points for treating the DAO (blue dots). DAO, depressor anguli oris.

improvement and maintained function. However, the patient should be informed that the effect duration will be shorter due to the reduced dose.

The *risorius* is rarely treated, only in cases of an asymmetrical smile. On smiling, the first skin fold formed laterally to the mouth corner is avoided, and 0.5 to 1 U of INCO is applied superficially.[37] In some cases of smile asymmetry, in which the lower teeth are exposed more on 1 side than the other, or in cases of unilateral idiopathic paralysis, the *depressor labii* can be treated independently. A dose of 1 to 2 U should be injected perpendicularly to the skin, 2 cm below the lower lip on the hyperkinetic side.[38]

The *mentalis* muscle elevates the lower lip and raises and wrinkles the skin of the chin, giving it an orange peel–like texture and a half-moon–shaped wrinkle above the chin, commonly referred to as a "witch's chin" (Figure 6, light blue dots). The treatment technique involves a deep intramuscular injection of 2 to 3 U in each bundle (2 points at the base of the chin) or a single dose in the center, massaging bilaterally. Caution should be taken to avoid accidentally treating the *depressor labii*, which is located laterally to the mentalis bundles.

If hyperactive, and/or with senescence, the *depressor anguli oris* muscle may induce the "sad mouth" look (downturned oral commissures). IncobotulinumtoxinA treatment can relax this muscle, allowing unopposed action of the mouth corner *levator* muscle. Injection should be administered at the end of a line connecting the nasolabial sulcus to the mandible or 1.5 cm medially to the anterior border of the *masseter* with the mandibular line (Figure 6; dark blue dots). The suggested dose ranges from 2 to 4 U per side.[39,40] The *depressor anguli oris* muscle partially covers the *depressor labii inferioris* muscle. When treating the former, the latter muscle may be affected, resulting in facial asymmetry and the inability to expose the lower teeth on the affected side.[40–42]

## Jawline and Neck

For the *masseter*, suggested doses may vary according to gender, ethnicity, patient demand, and other conditions such as bruxism.[4,43] Masseter hypertrophy gives the impression of heaviness and a square shape to the lower portion of the face, which is often considered aesthetically unpleasant by women as it represents characteristically male features. Genetic predisposition and chewing habits can contribute to the development of this appearance.[44] The relaxation of this muscle with INCO contributes to the slimming of the lower third of the face (Figure 7). While results can be observed after 15 days, the peak effect generally occurs at around 3 months due to muscle atrophy. The dose can range from 9 to 40 U per side, distributed in 3 to 5 injection points, avoiding the anterior border of the masseter (Figure 7; green dots). Dosage depends on the degree of muscular hypertrophy and the desired level of atrophy.[44]

Treatment of the *platysma* muscle has been used to improve facial contour or deep lower smile lines. The *platysma* functions as a major facial depressor, contributing to the emergence of horizontal wrinkles below the lower lip, oblique lines just below the mandibular border, and deep vertical wrinkles lateral to the oral commissures when smiling. The authors recommend treating the lower third of the face as a unit, similar to the approach for the *glabella* region.[45] The *platysma* treatment should include 2 points of deep intramuscular injection in the *mentalis*, along with 2 horizontal parallel lines of superficial injections on each side of the face (Figure 7; blue dots). The first line follows the contour of the mandible and consists of 3 points, spaced 2 cm apart, starting at the site of the *depressor anguli oris* and ending at the angle of the jaw. The second line is located 2 cm below the mandible and includes 4 injection points, spaced 2 cm apart, starting at the point between the mentalis and the *depressor anguli oris* and ending below the angle of the jaw. 2 U of INCO should be injected at each point, for a total of 14 U. In the lower line, the dose can be increased if necessary without the risk of facial asymmetries.[45] Superficial injection ensures restricted application to the *platysma* muscle. As a result, patients can expect improved mandibular contour, fewer oblique wrinkles below the mandible during *platysma* contraction, and fewer vertical wrinkles lateral and inferior to the oral commissure during smiling.[45]

## Skin Quality

Intradermal injection of BoNTAs to improve skin quality is currently under investigation. Because acetylcholine receptors can be found on many cell types (melanocytes,



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

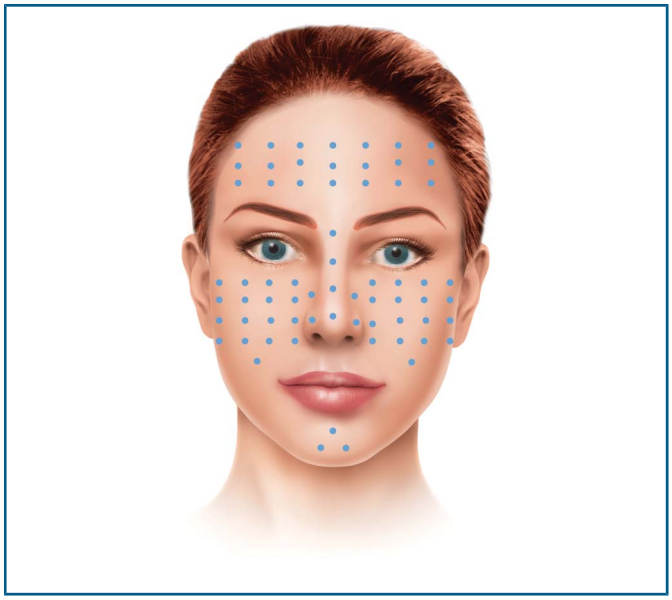


**Figure 8.** Injection protocol for INCO for skin quality. The lateral face, anterior medial cheek, forehead, chin, and nose must be injected intradermally, 0.5 U per point. The injection scheme can be adapted according to each patient's need. INCO, incobotulinumtoxinA.

keratinocytes, dermal fibroblasts, sebocytes, vascular endothelial cells), researchers theorize that the BoNTA mechanism of action may be utilized as a promising treatment for facial skin laxity, excessive sebum production, enlarged facial pores, facial redness, and to protect from UVB-induced hyperpigmentation.[46–49] However, due to the large quantity of DCs in the dermis, intradermal BoNTA injection may be more immunogenic than intramuscular or subcutaneous injection.[46] Therefore, the lower immunogenicity risk of INCO may be favorable during intradermal injection for skin quality indications.[50]

The reconstitution of a 100 U vial of INCO for skin quality treatments may vary between 2 mL (0.5 U/0.1 mL) and 5 mL (0.5 U/0.25 mL) of sterile saline solution.[46,49] In patients with thinner dermis, smaller volumes should be used to reduce spread and diminish the risk of paralyzing underlying muscles (Figure 8). For patients with a thicker dermis or extremely oily skin types, it may be advisable to use a higher dose per injection point. Using 1 U instead of 0.5 U could be beneficial in providing results for up to 4 months.

## Conclusion

In summary, the best outcomes of aesthetic INCO treatment occur through long-lasting muscle paresis, resulting in wrinkle effacement, facial contour improvement, and/or better eyebrow positioning. Despite the wide range of doses reported for each muscle, physicians should minimize adverse effects by utilizing the lowest INCO dosage that provides 4 months of efficacy; complete paralysis of facial muscle does not always provide the most natural result. Finally, correct injection depth is essential to target each muscle or structure and achieve optimal results.


## Acknowledgments

This is a company-funded submission based on a funded advisory board by Merz Aesthetics. Medical writing and editorial assistance were provided by Dr. Hélio Miot (Unesp, Botucatu-SP, Brazil) and Erin Scholz, PhD (Merz Aesthetics, Raleigh, NC) and was supported by Medical Affairs of Merz Aesthetics LATAM, through its affiliate Merz Pharmaceuticals, by Good Publication Practice guidelines.


## References


1. Frevert J. Pharmaceutical, biological, and clinical properties of botulinum neurotoxin type A products. *Drugs R D* 2015;15:1–9.
2. Park JY, Sunga O, Wanitphakdeedecha R, Frevert J. Neurotoxin impurities: a review of threats to efficacy. *Plast Reconstr Surg Glob Open* 2020;8:e2627.
3. Frevert J. Content of botulinum neurotoxin in Botox®/Vistabel®, Dysport®/Azzalure®, and Xeomin®/Bocouture®. *Drugs R D* 2010; 10:67–73.
4. Yutskovskaya Y, Gubanova E, Khrustaleva I, Atamanov V, et al. IncobotulinumtoxinA in aesthetics: Russian multidisciplinary expert consensus recommendations. *Clin Cosmet Investig Dermatol* 2015;8: 297–306.
5. Sundaram H, Huang PH, Hsu NJ, Huh CH, et al. Aesthetic applications of botulinum toxin A in Asians: an international, multidisciplinary, Pan-Asian consensus. *Plast Reconstr Surg Glob Open* 2016;4: e872.
6. Sundaram H, Signorini M, Liew S, Trindade de Almeida AR, et al. Global aesthetics consensus: botulinum toxin type A--evidence-based review, emerging concepts, and consensus recommendations for aesthetic use, including updates on complications. *Plast Reconstr Surg* 2016;137:518e–29e.
7. Ali S, Al Bukhari F, Al Nuaimi K, Elenany H, et al. Consensus statement on the use of botulinum neurotoxin in the middle east. *Clin Cosmet Investig Dermatol* 2023;16:2899–909.
8. Kerscher M, Wanitphakdeedecha R, Trindade de Almeida A, Maas C, et al. IncobotulinumtoxinA: a highly purified and precisely manufactured botulinum neurotoxin type A. *J Drugs Dermatol* 2019;18:52–7.
9. Carr WW, Jain N, Sublett JW. Immunogenicity of botulinum toxin formulations: potential therapeutic implications. *Adv Ther* 2021;38: 5046–64.
10. Ho WWS, Albrecht P, Calderon PE, Corduff N, et al. Emerging trends in botulinum neurotoxin A resistance: an international multidisciplinary review and consensus. *Plast Reconstr Surg Glob Open* 2022; 10:e4407.
11. Shi S, Zhu H, Xia X, Liang Z, et al. Vaccine adjuvants: understanding the structure and mechanism of adjuvanticity. *Vaccine* 2019;37: 3167–78.
12. Dressler D, Hefter H, Wohlfarth K, Baumer T. Botulinum neurotoxin: taking neutralizing antibodies seriously. *Curr Neurol.* 2014;8.
13. Dressler D. Therapeutically relevant features of botulinum toxin drugs. *Toxicon* 2020;175:64–8.
14. Curry A. Awareness of aesthetic and therapeutic crossover usage of botulinum neurotoxin type A. *Toxicon* 2024;237:107385–21.
15. Hefter H, Hartmann CJ, Kahlen U, Samadzadeh S, et al. Clinical improvement after treatment with IncobotulinumtoxinA (XEOMIN®) in patients with cervical dystonia resistant to botulinum toxin preparations containing complexing proteins. *Front Neurol* 2021;12: 636590.
16. Hefter H, Ürer B, Brauns R, Rosenthal D, et al. Significant long-lasting improvement after switch to incobotulinum toxin in cervical dystonia patients with secondary treatment failure. *Toxins (Basel)* 2022;14:44.
17. Carey WD. Incorrect reconstitution of incobotulinumtoxinA leads to loss of neurotoxin. *J Drugs Dermatol* 2014;13:735–8.
18. Hefter H, Brauns R, Ürer B, Rosenthal D, et al. Effective long-term treatment with incobotulinumtoxin (Xeomin®) without neutralizing antibody induction: a monocentric, cross-sectional study. *J Neurol* 2020;267:1340–7.





© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

19. Evidente VG, Truong D, Jankovic J, Comella CL, et al. IncobotulinumtoxinA (Xeomin®) injected for blepharospasm or cervical dystonia according to patient needs is well tolerated. *J Neurol Sci* 2014; 346:116–20.
20. Truong D. Botulinum neurotoxin and dystonia: a case for greater flexibility in treatment regimens. *CNS J* 2017;2:12–7.
21. Hefter H, Hartmann C, Kahlen U, Moll M, et al. Prospective analysis of neutralising antibody titres in secondary non-responders under continuous treatment with a botulinum toxin type A preparation free of complexing proteins--a single cohort 4-year follow-up study. *BMJ Open* 2012;2:e000646.
22. Carruthers A, Kane MAC, Flynn TC, Huang P, et al. The convergence of medicine and neurotoxins: a focus on botulinum toxin type A and its application in aesthetic medicine--a global, evidence-based botulinum toxin consensus education initiative: part I: botulinum toxin in clinical and cosmetic practice. *Dermatol Surg* 2013;39:493–509.
23. Polacco MA, Singleton AE, Barnes CH, Maas C, et al. A double-blind, randomized clinical trial to determine effects of increasing doses and dose-response relationship of IncobotulinumtoxinA in the treatment of glabellar rhytids. *Aesthet Surg J* 2021;41:NP500–511.
24. Thistle F. Greenhouse Gas Emissions and Botulinum Neurotoxin Packaging; 2012. [cited 2024 3/8/2024]. Avilable at: https://hospitalpharmacyeurope.com/news/editors-pick/greenhouse-gas-emissions-and-botulinum-neurotoxin-packaging/.
25. Lapatki BG, van Dijk JP, van de Warrenburg BPC, Zwarts MJ. Botulinum toxin has an increased effect when targeted toward the muscle's endplate zone: a high-density surface EMG guided study. *Clin Neurophysiol* 2011;122:1611–6.
26. Neubert J. Topographical Characterization of the Upper Facial Musculature Revealed by Means of High Density Surface Electromyography [dissertation]. University of Ulm; 2015.
27. de Sanctis Pecora C, Bussamara Pinheiro MV, Figueiredo V, Guerra R. Aesthetic ONE21 technique for Injecting IncobotulinumtoxinA into the forehead: initial experience with 86 patients. *J Drugs Dermatol* 2021;20:31–7.
28. de Sanctis Pecora C, Ventura Ferreira K, Amante Miot H. ONE21 technique for an individualized assessment and treatment of upper face wrinkles in five pairs of identical twins with IncobotulinumtoxinA. *J Cosmet Dermatol* 2022;21:1940–7.
29. de Sanctis Pecora C. One21: a novel, customizable injection protocol for treatment of the forehead with IncobotulinumtoxinA. *Clin Cosmet Investig Dermatol* 2020;13:127–36.
30. Abramo AC, Do Amaral TPA, Lessio BP, De Lima GA. Anatomy of forehead, glabellar, nasal and orbital muscles, and their correlation with distinctive patterns of skin lines on the upper third of the face: reviewing concepts. *Aesthetic Plast Surg* 2016;40:962–71.
31. Moqadam M, Frank K, Handayan C, Hakami M, et al. Understanding the shape of forehead lines. *J Drugs Dermatol* 2017;16:471–7.
32. da Cunha ALG, Vasconcelos R, Di Sessa D, Sampaio G, et al. IncobotulinumtoxinA for the treatment of glabella and forehead dynamic lines: a real-life longitudinal case series. *Clin Cosmet Investig Dermatol* 2023;16:697–704.
33. de Sanctis Pecora C, Pinheiro MVB, Ventura Ferreira K, Jacobino de Barros Nunes G, et al. The One21 technique: an individualized treatment for glabellar lines based on clinical and anatomical landmarks. *Clin Cosmet Investig Dermatol* 2021;14:97–105.
34. de Almeida AR, da Costa Marques ERM, Banegas R, Kadunc BV. Glabellar contraction patterns: a tool to optimize botulinum toxin treatment. *Dermatol Surg* 2012;38:1506–15.
35. Kane MA, Cox SE, Jones D, Lei X, et al. Heterogeneity of crow's feet line patterns in clinical trial subjects. *Dermatol Surg* 2015;41:447–56.
36. Tamura BM, Odo MY. Classification of periorbital wrinkles and treatment with botulinum toxin type A. *Surg Cosmet Dermatol* 2011; 3:129–34.
37. Bae JH, Choi DY, Lee JG, Seo KK, et al. The risorius muscle: anatomic considerations with reference to botulinum neurotoxin injection for masseteric hypertrophy. *Dermatol Surg* 2014;40:1334–9.
38. Benedetto AV. Asymmetrical smiles corrected by botulinum toxin serotype A. *Dermatol Surg* 2007;33:S32–6.
39. Boggio RF. Dynamic model of applied facial anatomy with emphasis on teaching of botulinum toxin A. *Plast Reconstr Surg Glob Open* 2017;5:e1525.
40. Trevidic P, Sykes J, Criollo-Lamilla G. Anatomy of the lower face and botulinum toxin injections. *Plast Reconstr Surg* 2015;136:84S–91S.
41. Wu DC, Fabi SG, Goldman MP. Neurotoxins: current concepts in cosmetic use on the face and neck--lower face. *Plast Reconstr Surg* 2015;136:76S–79S.
42. Landau M, Nestor MS, de Almeida AT, Al-Niaimi F. Botulinum toxin complications in registered and off-label aesthetic indications. *J Cosmet Dermatol* 2020;19:2484–90.
43. de Maio M, Wu WTL, Goodman GJ, Monheit G, et al. Facial assessment and injection guide for botulinum toxin and injectable hyaluronic acid fillers: focus on the lower face. *Plast Reconstr Surg* 2017; 140:393e–404e.
44. Angra K, Boen M, Alhaddad M, Fabi SG. Functional and aesthetic interplay between the platysma and masseter muscles. *Dermatol Surg* 2020;46:719–20.
45. de Almeida ART, Romiti A, Carruthers JDA. The facial platysma and its underappreciated role in lower face dynamics and contour. *Dermatol Surg* 2017;43:1042–9.
46. Park JY, Cho SI, Hur K, Lee DH. Intradermal microdroplet injection of diluted Incobotulinumtoxin-A for sebum control, face lifting, and pore size improvement. *J Drugs Dermatol* 2021;20:49–54.
47. Vachiramon V, Kositkuljorn C, Leerunyakul K, Rattananukrom T, et al. A study of botulinum toxin A for ultraviolet-induced hyperpigmentation: a randomized controlled trial. *Dermatol Surg* 2021;47: e174–8.
48. Oh SH, Lee Y, Seo YJ, Lee JH, et al. The potential effect of botulinum toxin type A on human dermal fibroblasts: an in vitro study. *Dermatol Surg* 2012;38:1689–94.
49. Dayan SH, Pritzker RN, Arkins JP. A new treatment regimen for rosacea: onabotulinumtoxinA. *J Drugs Dermatol* 2012;11:e76–9.
50. Zhang L, Wang W, Wang S. Effect of vaccine administration modality on immunogenicity and efficacy. *Expert Rev Vaccines* 2015;14: 1509–23.



© 2024 by the American Society for Dermatologic Surgery, Inc. Published by Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.